

Potential Science Fair Projects

Assistive spatial perception for blind and low-vision (BLV) users. Top three impact-first ideas. June 2026.

Each project is framed for an ISEF Engineering / applied-AI entry, in the standard **Pitch / Litmus Test (Novel, Feasible, Impactful, Testable)** structure. All three run on a commodity phone and evaluate on existing public benchmarks plus a small user study, and all three also support an optional head-mounted wearable build using the Snap Spectacles camera with spatial audio (a head-mounted form factor BLV users tend to prefer), with XREAL as a possible alternative. They sit in the assistive / accessibility impact domain that historically places well at ISEF. Target numbers are concrete.

Project Idea 1: “Which Way Am I Facing?”, Spatial-Orientation Assistance for Blind Users

Reframes AI scene understanding into an assistive aid for the hardest question a blind traveler faces, which way am I oriented and where are things relative to me, the reasoning that underlies safe independent travel.

The Pitch: Blind and low-vision users can already identify objects with existing apps, but still struggle with spatial orientation, the egocentric “which way am I facing, where is the door relative to me, which way do I turn” reasoning that underlies independent mobility. On EgoBlind, the first egocentric video-QA benchmark collected from blind people (111 blind contributors), the best multimodal model (GPT-4o) reaches only 59.3% accuracy versus 87.4% for humans, and the authors single out spatial orientation as an “extreme challenge to all MLLMs.” This project builds a method that improves spatial-orientation question answering from a blind user’s first-person video, then validates it with blind and low-vision participants. The contribution is a targeted method for the least-solved, highest-stakes sub-task of assistive perception, not a general “describe the scene” tool.

Litmus Test:

- **NOVEL:** It targets the specific sub-task that published benchmarks show is least solved, rather than restating the general blind-assistance problem. It differs from general assistive visual-QA and from navigation systems such as NaviNote (CHI 2026), which do route localization and last-meters guidance rather than egocentric spatial-orientation reasoning. The novelty is a method contribution on a named, quantified open gap.
 - **FEASIBLE:** Software-first and solo-feasible. The EgoBlind dataset, the baseline numbers, and the evaluation metric already exist publicly, so the core work is method design (fine-tuning, prompting, or a spatial-reasoning module) plus a small user study. It runs on a phone, and an optional head-mounted demo can use the Spectacles camera with voice output.
 - **IMPACTFUL:** Spatial orientation underlies safe independent travel for tens of millions of blind people, so closing even part of the 28-point human-AI gap is a direct independence and safety benefit. The assistive / translational framing is a proven ISEF-winning impact lane, as seen in the mixed-reality dementia-therapy and EEG-prosthesis grand-award winners.
 - **TESTABLE: Testable Goal:** “To develop a method that improves spatial-orientation question-answering accuracy on the EgoBlind benchmark by at least 10 absolute percentage points over the best published baseline (GPT-4o, 59.3%), and to show in a study with at least 8 blind or low-vision participants that its spatial-orientation answers are rated significantly more accurate and more helpful than the baseline ($p < 0.05$).”
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Project Idea 2: Overhead-Hazard Detection, Catching the Obstacles a White Cane Cannot

Reframes obstacle detection around the documented blind spot of existing mobility aids, the head- and torso-height hazards that a white cane physically cannot reach.

The Pitch: White canes and most assistive apps detect ground-level obstacles but miss head- and torso-height hazards such as low branches, protruding signage, open cabinet doors, and side mirrors, a documented and common source of injury for blind travelers. A systematic review of 646 studies names aerial-obstacle detection as a high-value but under-researched task that researchers have largely neglected. This project builds a real-time detector that identifies overhead and torso-height obstacles from a chest- or head-mounted camera and warns the user with directional spatial audio, then tests whether it reduces head-height collisions on a controlled course. The contribution is an assistive perception system aimed squarely at a hazard class existing aids ignore.

Litmus Test:

- **NOVEL:** It directly targets a task the literature explicitly flags as under-served, rather than ground-plane obstacle avoidance or general scene description. The novelty lives in the task focus plus the directional spatial-audio warning design rather than in a brand-new architecture, and the under-explored status means low risk of being scooped.
- **FEASIBLE:** It is buildable with a phone or a head-mounted camera plus near-real-time detection. It can bootstrap from existing detection models fine-tuned on a curated set of overhead-hazard imagery, and a controlled obstacle course makes the user study tractable for a solo researcher. The Spectacles camera is a natural head-level mounting point for the wearable build.
- **IMPACTFUL:** It prevents real, recurring head and face injuries, a concrete and vivid human benefit judges grasp instantly, and it addresses a gap the field itself acknowledges it neglects, which strengthens the impact narrative.
- **TESTABLE: Testable Goal:** “To build a real-time overhead-hazard warning system (≥ 10 fps on a phone or head-mounted camera) that detects head-height obstacles with at least 85% recall at a low false-alarm rate, and to demonstrate in a controlled-course study with at least 10 blindfolded or BLV participants a statistically significant reduction in head-height contacts versus cane-only travel ($p < 0.05$).”

Project Idea 3: Real-Time On-Device Blind-Assist Perception, Because a Slow Answer Is an Unsafe Answer

Reframes assistive AI as a latency and deployability problem, because for a blind traveler in motion a delayed or connectivity-dependent answer is a safety failure, not just an inconvenience.

The Pitch: Most capable assistive-perception systems run in the cloud, which adds latency, requires connectivity, and streams a blind user's continuous camera feed off-device, a real privacy concern. For a person in motion, a one-to-two-second delay can mean a missed hazard. This project builds an on-device pipeline that performs assistive scene and hazard perception in real time on a commodity smartphone (with an optional Spectacles wearable build), and rigorously benchmarks the on-device-versus-cloud tradeoff on latency, accuracy, energy, and task success. The contribution is a deployable, private, offline-capable assistive system plus a quantified characterization of what real-time on-device assistance actually costs and delivers.

Litmus Test:

- **NOVEL:** Most blind-assistance systems are cloud- or lab-bound, so a quantified on-device real-time characterization for assistive perception is a defensible systems contribution. It differs from accuracy-only benchmarks by treating latency and energy as *safety* variables rather than afterthoughts.
- **FEASIBLE:** It needs only software plus a phone (with the Spectacles as an optional wearable target), and it is solo-feasible. It leverages compression and quantization of existing detectors, and the benchmark half front-loads heavy quantified data with no human subjects, de-risking the timeline before any user study.
- **IMPACTFUL:** Deployability, offline use, privacy, and real-time response are the actual barriers to real-world adoption of assistive AI, so closing them is high-impact and yields the artifact-plus-impact combination ISEF rewards.
- **TESTABLE: Testable Goal:** “To build an on-device assistive-perception pipeline that runs at ≥ 15 fps with under 150 ms end-to-end latency on a commodity smartphone while retaining at least 90% of the cloud model’s task accuracy, and to quantify the latency-accuracy-energy frontier across on-device versus cloud, including runtime-to-battery-depletion.”

All three sit in the blind / low-vision assistive-perception space, the strongest combination of high ISEF impact, solo feasibility, and a well-evidenced open gap that the research surfaced. Key risk to manage, this area is moving fast (EgoBlind, GuideDog, mmWalk, and NaviNote all appeared within about 12 months), so each project must stay on a narrow, under-served sub-task rather than the general problem. Any human-participant study needs IRB / SRC pre-approval before recruitment, and recruiting adults (18+) simplifies that review.